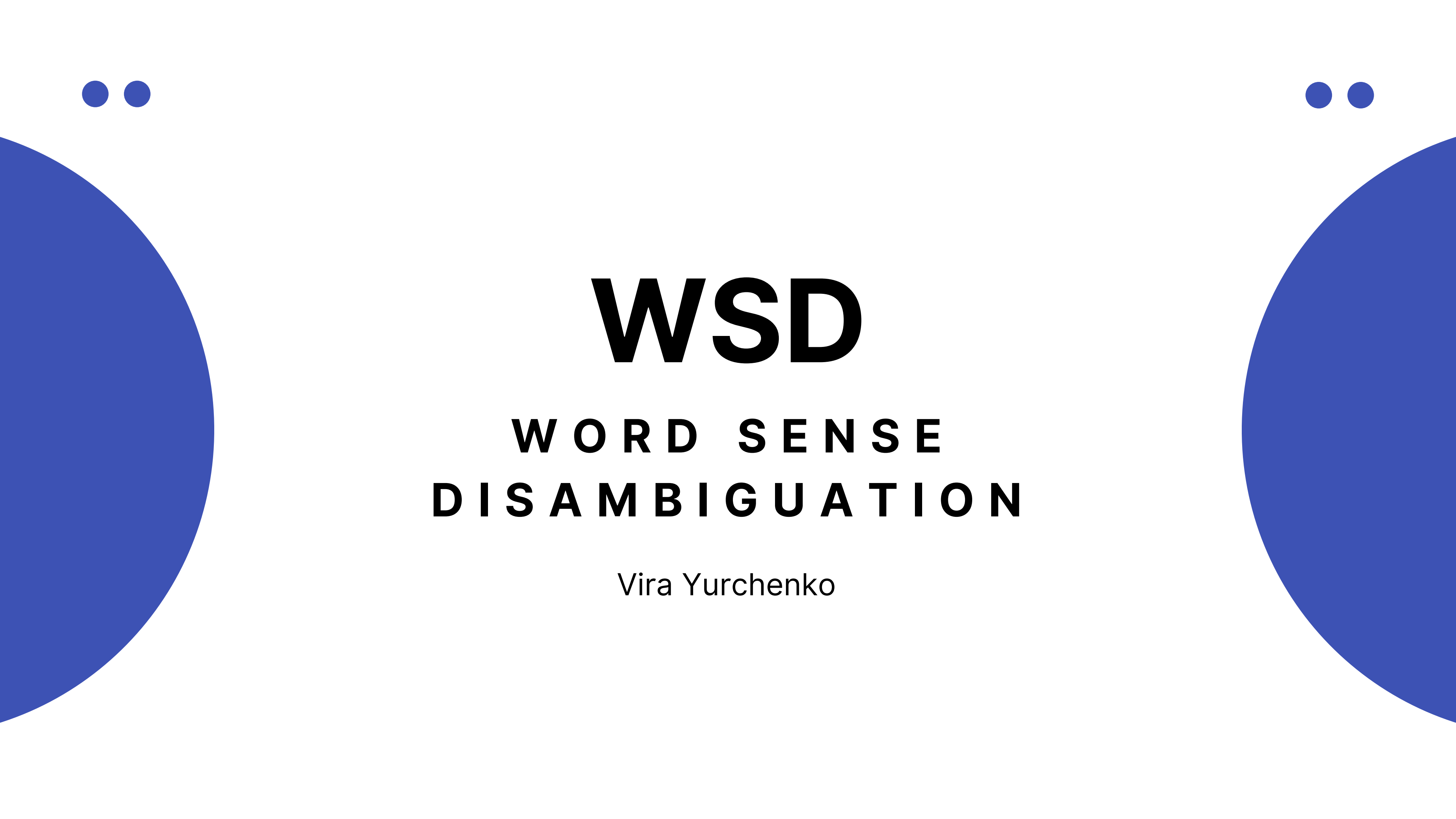


The slide features a white background with decorative elements in blue. In the top-left and top-right corners, there are pairs of small blue dots. On the left and right sides, there are large blue semi-circles that appear to be part of a larger circle cut off by the edge of the slide.

WSD

**WORD SENSE
DISAMBIGUATION**

Vira Yurchenko

THE PURPOSE OF THIS PROJECT

- It assign correct WordNet senses to lemmas in a JSON corpus and tests how different context sizes affect disambiguation quality using Ollama and WordNet NTUMC.
- Train AI to understand the meaning of a word in a specific sentence and select the correct ID from the WordNet (or special tags if the word is not there: names, places, errors).



IMPROVEMENTS

- Change the WordNet (Open English WordNet → WordNet NTUMC)
- Rewrite the prompt
- Compare context's sizes:

What is better for understanding the meaning of the word?

(3) sentences - Word OR (3) sentences - Word - (3) sentences.



PROBLEMS

- Connection reset (code crashed in the middle because of loss wifi connection).

Start to use tmux. The terminal multiplexer that allows to run the code in background sessions and close the laptop while the server was doing its work.

- OOM (Out of Memory, status code: 500). The server killed the process because of lack of memory while trying to run the bigger models (14B..).

RESULTS

Model	Context	Total Accuracy	Macro Precision	Micro Precision
Gemma 2:9b	0	46.41%	0.433	0.662
	1	44.44%	0.504	0.639
	2	44.65%	0.447	0.637
	3	43.66%	0.432	0.627
Llama 3.1:8b	0	43.31%	0.413	0.596
	1	42.54%	0.419	0.573
	2	42.89%	0.417	0.565
	3	41.90%	0.417	0.561
Qwen 2.5:7b	0	41.90%	0.398	0.592
	1	39.93%	0.423	0.565
	2	39.15%	0.418	0.563
	3	39.79%	0.387	0.571
Mistral-nemo:12	0	35.56%	0.363	0.481
	1	35.21%	0.367	0.477
	2	35.28%	0.350	0.469
	3	35.00%	0.356	0.464

(3) sentences - word

Model	Context	Total Accuracy	Macro Precision	Micro Precision
Gemma 2:9b	0	49.23%	0.510	0.688
	1	48.59%	0.534	0.688
	2	48.59%	0.533	0.684
	3	48.94%	0.530	0.687
Llama 3.1:8b	0	45.35%	0.474	0.649
	1	44.01%	0.487	0.644
	2	45.42%	0.513	0.663
	3	45.35%	0.512	0.659
Qwen 2.5:7b	0	43.66%	0.471	0.625
	1	42.68%	0.551	0.620
	2	42.04%	0.517	0.619
	3	42.25%	0.566	0.612
Mistral-nemo:12	0	49.65%	0.488	0.691
	1	49.72%	0.474	0.685
	2	49.37%	0.558	0.673
	3	50.07%	0.598	0.685

(3) sentences - word - (3) sentences

FUTURE WORK

- Try to run bigger models (if it is possible)
- Use the new server
- Improve the code



**THANK YOU FOR
YOUR ATTENTION!**

